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Prepare the system for future work

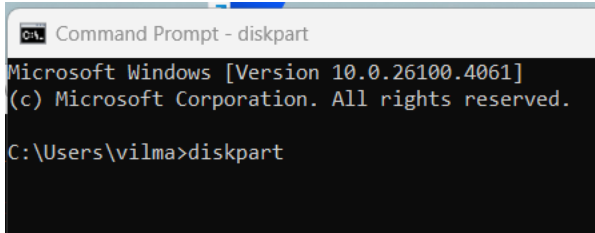
Raspberry Pi documentation:

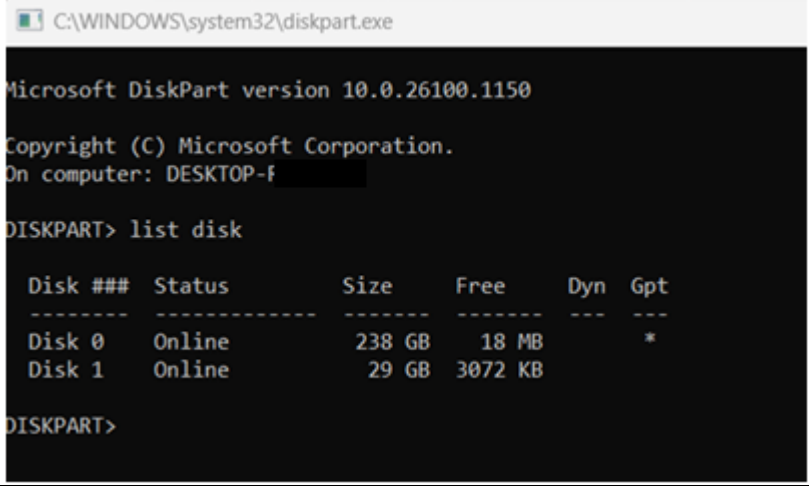
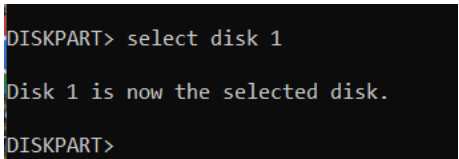
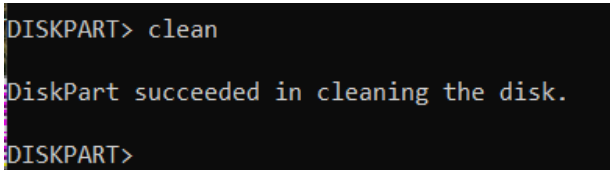
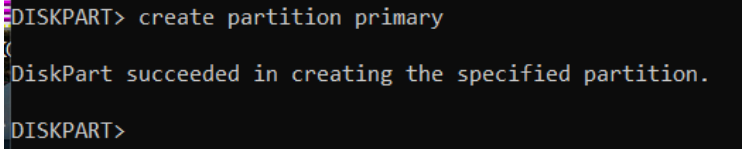
<https://www.raspberrypi.com/documentation/computers/getting-started.html#setting-up-your-raspberry-pi>

Preparing the SD card

At first, you need to format the SD card. You will do this on the lab's computer with Windows OS (or your own personal computer).

Open Command Prompt (Windows: cmd) and use the following commands to format the SD card:

1.	diskpart	“The diskpart command interpreter helps you manage your computer's drives (disks, partitions, volumes, or virtual hard disks). Before you can use diskpart commands, you must first list, and then select an object to give it focus. After an object has focus, any diskpart commands that you type will act on that object.” https://learn.microsoft.com/en-us/windows-server/administration/windows-commands/diskpart  The screenshot shows a Windows Command Prompt window titled 'Command Prompt - diskpart'. It displays the Windows version '10.0.26100.4061' and the user 'vilma' at the prompt 'C:\Users\vilma>diskpart'.
2.	list disk	“list - Displays a list of disks, of partitions in a disk, of volumes in a disk, or of virtual hard disks (VHDs).” Usage: list disk list volume list partition list vdisk

		 <pre> C:\WINDOWS\system32\diskpart.exe Microsoft DiskPart version 10.0.26100.1150 Copyright (C) Microsoft Corporation. On computer: DESKTOP-F... DISKPART> list disk Disk ### Status Size Free Dyn Gpt ----- - Disk 0 Online 238 GB 18 MB Disk 1 Online 29 GB 3072 KB DISKPART> </pre>
3.	Select disk 'x'	“select - Shifts the focus to a disk, partition, volume, or virtual hard disk (VHD).” Usage: select disk 1 select disk 0 You need to select the SD card (32 GB).  <pre> DISKPART> select disk 1 Disk 1 is now the selected disk. DISKPART> </pre>
4.	clean	“clean - Removes any and all partition or volume formatting from the disk with focus.”  <pre> DISKPART> clean DiskPart succeeded in cleaning the disk. DISKPART> </pre>
5.	create partition primary	“create - Creates a partition on a disk, a volume on one or more disks, or a virtual hard disk (VHD). A primary partition is where an Operating System can be installed.”  <pre> DISKPART> create partition primary DiskPart succeeded in creating the specified partition. DISKPART> </pre>
6.	format fs=fat32 quick	“format - Formats a disk to accept files. fs - Specifies the type of file system.

		quick - Performs a quick format.” <pre>DISKPART> format fs=fat32 quick 100 percent completed DiskPart successfully formatted the volume. DISKPART></pre>
7.	assign	“assign - Assigns a drive letter or mount point to the volume with focus.” <pre>DISKPART> assign DiskPart successfully assigned the drive letter or mount point. DISKPART></pre>

Installing the operating system

There are several different Raspberry Pi models used in the laboratory (Raspberry Pi 3B, Raspberry Pi 4B, and Raspberry Pi 5). In addition, laboratory sessions are held in subgroups. Therefore, the single-board computers will be shared among students from different subgroups. Consequently, the system must be configured to ensure that it is accessible to all users.

The installation process is described in the Raspberry Pi documentation. We will use Raspberry Pi Imager, available from the official Raspberry Pi website, to write the operating system image to an SD card.

We will use the same username on all Raspberry Pi computers. This refers to the Linux user account on the Raspberry Pi, not the user account on the Raspberry Pi website.

Create User

username: pi313

password: berryRasp

Wireless LAN settings

SSID: FF313

Password: [ff313#lab]

hostname: raspberrypi.local

OpenCV installation

1. Flash the latest Raspberry Pi OS (Trixie) to an SD card. This should be the first option available in Raspberry Pi Imager. (You have already done this!)

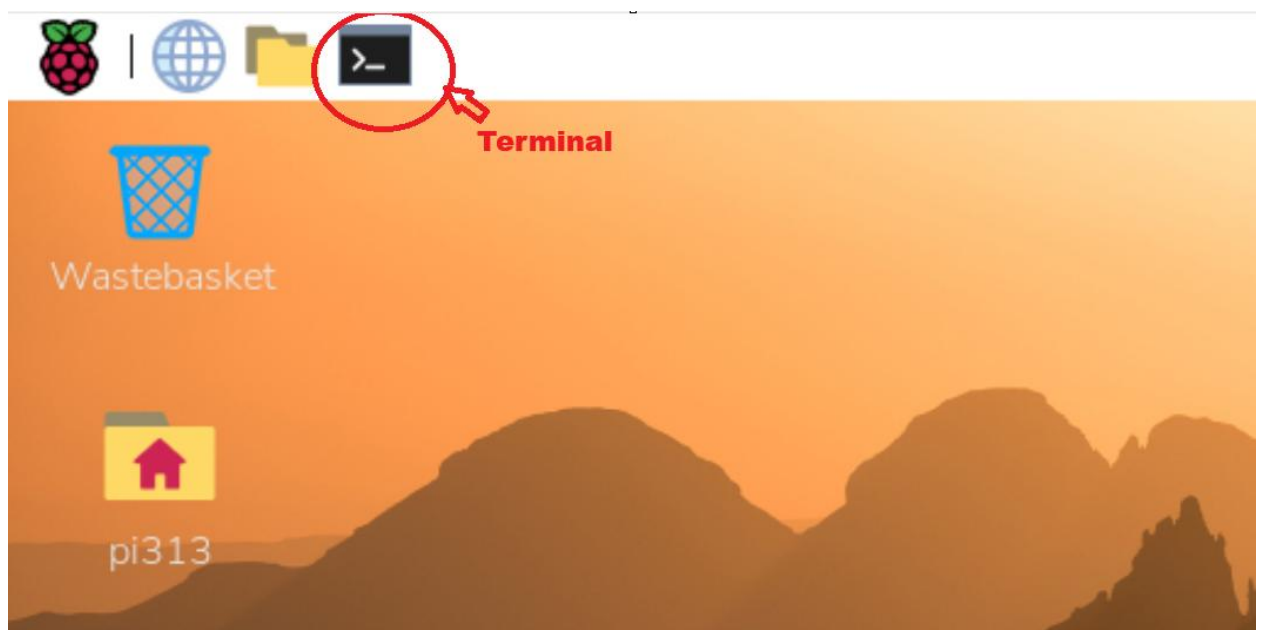
2. Boot the Raspberry Pi and install any available updates by running:

```
sudo apt update && sudo apt full-upgrade -y
```

3. Once the updates are complete, reboot the Raspberry Pi.

4. Install OpenCV by running:

```
sudo apt install python3-opencv
```



Remotely connect to your Raspberry Pi

Raspberry Pi Connect allows you to remotely access and control your Raspberry Pi from a web browser.

<https://www.raspberrypi.com/software/connect/>

		e-mail address	RPi ID password
1.	RPi-01	rubus01@tutamail.com	shepherd01??
2.	RPi-02	avite02@tutamail.com	schnauzer02??
3.	RPi-03	framboise03@mailfence.com	poodle03??
4.	RPi-04	framboise04@tutamail.com	husky04??
5.	RPi-05	framboise05@proton.me	terrier05??
6.	RPi-06	lampone06@mailfence.com	retriever06??
7.	RPi-07	rpi07@tutamail.com	buldogas07??
8.	RPi08	rpi313lab@gmail.com	woody08%%